Sentiment Analysis of IMDB Movie Reviews

A Comparative Study of Machine Learning, Deep Learning, and Transformer Models

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Abstract

This report presents a comprehensive study on sentiment analysis applied to the IMDB Movie Reviews dataset, which contains 50,000 labeled reviews. The primary objective is to classify reviews as positive or negative using three families of models: classical machine learning (Logistic Regression, Linear SVC, Naive Bayes), recurrent deep learning (Simple RNN and LSTM), and a pre-trained transformer (DistilBERT). Each model family is paired with a tailored preprocessing pipeline aggressive for TF-IDF, light for sequence models, and minimal for the transformer. Linear SVC achieved the highest accuracy among classical models at 91%, while DistilBERT achieved the best weighted F1-score of 91% among all models with superior recall on positive reviews. The LSTM matched Logistic Regression at 90% accuracy, outperforming the Simple RNN significantly.

1. Introduction

Sentiment analysis, a core task in Natural Language Processing (NLP), involves classifying text by its expressed opinion, typically positive or negative. Movie review platforms generate millions of opinionated texts daily, and automatically identifying their sentiment enables applications ranging from recommendation systems to brand monitoring.

The IMDB dataset is a standard benchmark in NLP, consisting of 25,000 positive and 25,000 negative reviews. This project explores how different model families from simple bag-of-words classifiers to state-of-the-art transformers compare on this task, and investigates the preprocessing strategies best suited to each.

1.1 Objectives

- Design and compare three preprocessing pipelines tailored to ML, DL, and Transformer models.
- Train and evaluate five models across three model families.
- Analyze trade-offs between model complexity, training cost, and performance.
- Identify the best model for production deployment on this task.

2. Dataset

The dataset used is the IMDB Dataset of 50K Movie Reviews, publicly available on Kaggle. It is one of the most widely cited benchmarks for binary sentiment classification.

Attribute	Details
Total Reviews	50,000
Positive Reviews	25,000 (50%)
Negative Reviews	25,000 (50%)
Duplicates Removed	418 exact duplicate reviews removed
Final Dataset Size	49,582 reviews after deduplication
Train / Test Split	80% / 20% (39,666 train | 9,917 test), stratified
Avg. Review Length	~230 words

95th Pct. Length	580 words (used as max sequence length for DL)

2.1 Key EDA Findings

- The dataset is perfectly balanced 50% positive, 50% negative, so accuracy is a fair metric.
- Review lengths vary widely: the median is ~160 words but the 99th percentile exceeds 1,600 words.
- Positive reviews tend to be slightly longer on average than negative reviews.
- Top positive words: film, great, story, character, performance, love, best, beautiful.
- Top negative words: bad, worst, boring, waste, awful, terrible, poor, nothing.
- Bigram analysis confirms strong polarizing patterns: ‘highly recommend’, ‘must see’ (positive) vs ‘waste time’, ‘bad acting’ (negative).
- Top positive trigrams: ‘one best films’, ‘one best movies’, ‘well worth watching’; top negative trigrams: ‘worst movie ever’, ‘waste time money’, ‘bad acting script’.
- Vocabulary overlap analysis: ~85,000 unique words in positive reviews and ~83,000 in negative; ~75,000 are shared across both classes, with ~10,000 exclusive to positive and ~8,000 exclusive to negative.
- Review length correlates weakly with sentiment: longer reviews (>500 words) trend more positive (enthusiastic viewers write more), while very short reviews (<100 words) are near-balanced at ~50% positive.

2.2 Text Length & Vocabulary Analysis

Four length-related features were computed after stripping HTML and URLs: word count, character count, sentence count (punctuation-based), and average word length. Their distributions are summarized below.

- **Word Count:** Mean ~233, Median ~163, 75th pct ~299, 90th pct ~461, 95th pct ~580, 99th pct >1,600. The distribution is heavily right-skewed; the bulk of reviews fall in the 100–300 word range.
- **Character Count:** Mirrors the word count pattern with a median of ~900 characters. Positive reviews exhibit a slightly higher median character count than negative reviews, consistent with the word count finding.
- **Sentence Count:** Estimated via punctuation markers (. ! ?). Positive reviews average more sentences per review, suggesting more structured and elaborated writing compared to negative reviews.
- **Average Word Length:** Approximately 4.8–5.0 characters per word across both classes. This metric shows very little sentiment discrimination, indicating vocabulary sophistication (word length) is not a strong differentiator.

2.3 Vocabulary Overlap

A vocabulary overlap analysis was performed on post-cleaned, stop-word-removed tokens (minimum 3 characters). The results quantify how much lexical overlap exists between the two sentiment classes.

- **Positive vocabulary:** ~85,000 unique words (after stop-word removal, min length 3).
- **Negative vocabulary:** ~83,000 unique words.

- **Shared words:** ~75,000 words appear in both classes — the high overlap confirms that sentiment is carried by frequency and context rather than by lexical exclusivity.
- **Exclusive to positive:** ~10,000 words (e.g., praise and admiration-specific terminology).
- **Exclusive to negative:** ~8,000 words (e.g., criticism-specific and pejorative terminology). The smaller exclusive-negative vocabulary reinforces that negative sentiment is more concentrated around a core set of strong terms.

2.4 Review Length vs. Sentiment Heatmap

Reviews were bucketed into seven word-count ranges and the positive-class percentage was computed per bucket. The stacked bar chart reveals a notable monotonic trend: the proportion of positive reviews increases with review length.

- **<100 words:** ~50% positive — near-balanced; short reviews show no sentiment bias.
- **200–500 words:** ~53–57% positive; mid-length reviews begin to skew positive.
- **700+ words:** ~60–65% positive; long reviews are predominantly positive, consistent with the behavioural pattern that enthusiastic viewers invest more effort in their write-ups.

Note: This length-sentiment correlation is a property of the raw data and does not impact the 80/20 stratified split, which preserves the 50/50 class ratio in both train and test sets.

3. Preprocessing

Three separate preprocessing pipelines were implemented, each optimized for the model family it feeds. A universal cleaning layer (HTML removal, URL removal, bracket stripping) is shared across all pipelines.

Pipeline	Lowercase	Stopwords Removed	Lemmatized	Notes
ML (TF-IDF)	Yes	Yes*	Yes (POS-aware)	*Negations preserved
DL Sequence	Yes	No	Yes (POS-aware)	Context words kept for RNN/LSTM
Transformer	No	No	No	WordPiece handles subwords

3.1 ML Preprocessing Pipeline

The ML pipeline applies the most aggressive transformations to maximize the effectiveness of TF-IDF sparse representations. Steps are applied in the following order:

- Remove HTML tags using BeautifulSoup.
- Remove square bracket content (e.g., [spoiler]).
- Remove URLs.
- Convert to lowercase.
- Expand contractions (e.g., don't → do not) using a curated 35-entry dictionary.
- Remove special characters and normalize whitespace.
- Tokenize using NLTK ToktokTokenizer.

- Remove stop words, preserving 22 negation words (not, never, can't, etc.) critical for sentiment polarity.
- POS-aware lemmatization using WordNetLemmatizer, mapping Penn Treebank POS tags to WordNet constants.

3.2 DL Sequence Pipeline (RNN / LSTM)

The DL sequence pipeline applies light preprocessing. Stop words are intentionally preserved because the embedding layer and recurrent units benefit from full contextual sequences. The pipeline omits stop-word removal but keeps POS-aware lemmatization.

3.3 Transformer Pipeline (DistilBERT)

The transformer pipeline is minimal; only HTML tags, URLs, and extra whitespace are removed. Casing, contractions, and stop words are preserved because DistilBERT's WordPiece tokenizer was pre-trained on raw text and handles subword segmentation internally. Aggressive preprocessing would discard information that the model was specifically trained to use.

4. Modeling

4.1 Machine Learning Models

Classical ML models were trained on TF-IDF vectorized text using unigrams and bigrams (ngram_range=(1,2)). Two vocabulary sizes were compared: 10,000 and 50,000 features. Results reported below use 50k features, which consistently outperformed the 10k configuration across all three models.

4.1.1 Logistic Regression

Logistic Regression with L2 regularization (max_iter=1000). A well-established baseline for text classification that models the log-odds of the positive class as a linear combination of TF-IDF features.

4.1.2 Linear SVC

Linear Support Vector Classifier finds the maximum-margin hyperplane in the TF-IDF feature space. Particularly effective for high-dimensional sparse text representations. Best-performing classical model.

4.1.3 Naive Bayes (Multinomial)

Multinomial Naive Bayes models word frequency distributions per class using Bayes' theorem. Computationally, the fastest model, but its conditional independence assumption limits performance on reviews with complex phrasing.

4.2 Deep Learning Models

Both DL models use a shared tokenization and padding pipeline. A Keras Tokenizer was fit on the training set with vocab_size=50,000 and oov_token='<OOV>'. Sequences were padded/truncated to max_length=580 (the 95th percentile of training review lengths). An EarlyStopping callback (patience=3, restore_best_weights=True) was used to prevent overfitting.

4.2.1 Simple RNN

Architecture: Embedding(50000, 64, mask_zero=True) → SimpleRNN(64) → Dense(16, relu) → Dropout(0.1) → Dense(1, sigmoid). Trained for up to 15 epochs, batch_size=64, Adam(lr=0.001). Simple RNNs suffer from vanishing gradients on long sequences, which explains the weaker performance compared to LSTM.

4.2.2 LSTM

Architecture: Embedding(50000, 64, mask_zero=True) → LSTM(64) → Dense(32, relu) → Dropout(0.3) → Dense(1, sigmoid). Trained for up to 10 epochs, batch_size=64, Adam(lr=0.0001). LSTM gates (input, forget, output) address the vanishing gradient problem, enabling the model to capture long-range dependencies in reviews — the deciding factor in its superior performance over Simple RNN.

4.3 Transformer — DistilBERT Fine-Tuning

DistilBERT (distilbert-base-uncased) is a distilled version of BERT, retaining 97% of its language understanding while being 40% smaller and 60% faster. The model was fine-tuned end-to-end on the IMDB training set using the HuggingFace Trainer API.

Key hyperparameters:

- Max sequence length: 256 tokens (covers ~90th percentile, halves memory vs. 512).
- Epochs: 8 (with EarlyStopping on eval_loss).
- Learning rate: 2e-5 (standard BERT fine-tuning range: 1e-5 to 5e-5).
- Batch size: 64 per device.
- Weight decay: 0.05.
- Warmup ratio: 0.1 of total steps.
- Mixed precision (fp16) enabled when GPU is available.
- Dynamic padding via DataCollatorWithPadding for memory efficiency.

5. Results & Evaluation

All models were evaluated on the same held-out test set of 9,917 reviews (4,940 negative, 4,977 positive). The following table summarizes classification performance.

Model	Accuracy	Precision	Recall	F1-Score	Notes
Logistic Regression	90%	90%	90%	0.90	TF-IDF 50k
Linear SVC	91%	91%	91%	0.91	TF-IDF 50k — Best ML
Naive Bayes	88%	88%	88%	0.88	TF-IDF 50k
Simple RNN	83%	83%	83%	0.82	Weakest model

LSTM	90%	90%	90%	0.90	Best DL sequence model
DistilBERT	**90%**	**91%**	**90%**	**0.91**	**Best overall — highest recall on positive (94%)**

5.1 Key Observations

- Linear SVC is the strongest classical model (91% accuracy), and trains in seconds making it highly practical for production.
- LSTM significantly outperforms Simple RNN (90% vs 83%), demonstrating the importance of gating mechanisms for long-text sentiment.
- DistilBERT achieves the highest positive-class recall (94%) it is the best model for minimizing false negatives on positive reviews.
- Simple RNN is the only model that falls below 85%, making it unsuitable for this task without architectural improvements.
- Increasing TF-IDF features from 10k to 50k consistently improved ML model performance, indicating rich vocabulary coverage matters.
- The transformer did not dramatically outperform Linear SVC in accuracy (90% vs 91%), but its recall advantage on positives is meaningful in asymmetric-cost scenarios.

5.2 Confusion Matrix Analysis

Analyzing the confusion matrices reveals that all models make more errors on negative reviews (more false positives) than on positive ones a common pattern in IMDB data due to the prevalence of mixed-sentiment language in critical reviews. DistilBERT is the notable exception: it achieves 94% recall on positive reviews by learning subtle linguistic cues that simpler models miss.

6. Discussion

6.1 Model Selection Recommendations

The right model depends on the deployment context:

- Production (latency-sensitive): Linear SVC 91% accuracy, trains in <30 seconds, inference in milliseconds.
- Production (quality-sensitive): DistilBERT best positive recall, handles nuanced language and negation naturally.
- Baseline / interpretability: Logistic Regression coefficients are directly interpretable as feature importances.
- Do not deploy Simple RNN: its 83% accuracy is significantly below alternatives with similar training complexity.

6.2 Effect of Preprocessing Choices

- Preserving negation words in the ML stopword filter is critical removing 'not' would flip the polarity of hundreds of phrases.

- POS-aware lemmatization provides more accurate root forms than stemming (e.g., 'better' → 'good' as adjective, not 'bett').
- Keeping stop words for DL sequence models preserves the grammatical structure that RNNs and LSTMs rely on for contextual encoding.
- The transformer preprocessing is deliberately minimal DistilBERT's pre-training on raw text means any additional cleaning risks removing information the model was trained to leverage.

6.3 Limitations

- Results are specific to the IMDB domain; sentiment models often require domain adaptation for other review types.
- The Simple RNN's weakness may be addressable with bidirectional architecture or attention mechanisms.
- DistilBERT was fine-tuned for 8 epochs, which may risk overfitting 3 epochs is the standard recommendation for BERT-family models.
- No hyperparameter search was performed

7. Conclusion

This project demonstrates that effective sentiment analysis on IMDB reviews can be achieved across a spectrum of model families, each with distinct trade-offs. Classical models with TF-IDF particularly Linear SVC remain highly competitive and are orders of magnitude cheaper to train than deep models. The LSTM substantially improves over Simple RNN, confirming the value of gated architectures for long-form text. DistilBERT represents the ceiling of performance achievable on this dataset with standard fine-tuning, particularly for recall on positive reviews.

The preprocessing pipeline design proved as important as model selection: tailoring cleaning, tokenization, and feature representation to each model family was essential to achieving competitive results. Future work could explore ensemble methods combining TF-IDF classifiers with transformer representations, data augmentation for harder examples, and cross-domain evaluation.

References

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